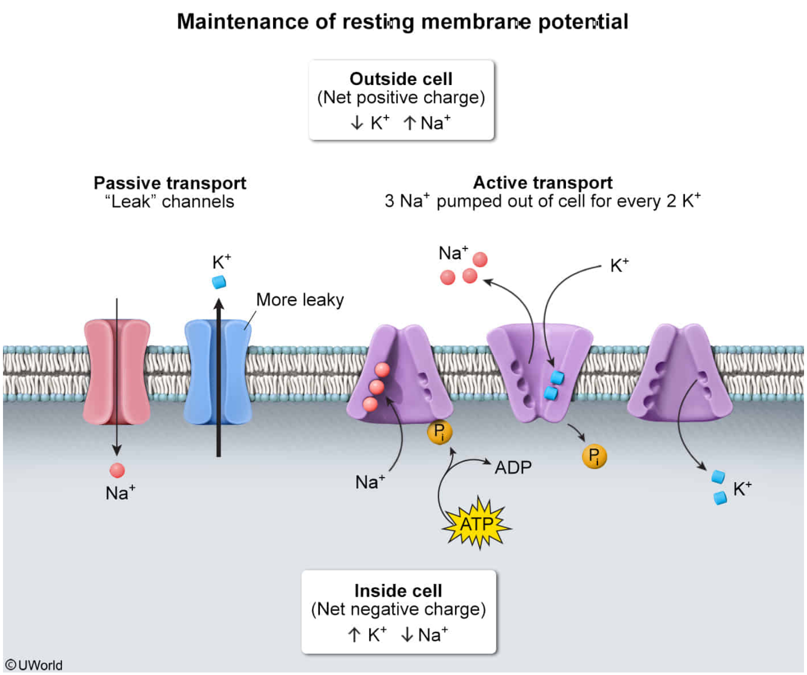
Which of the following helps maintain the resting membrane potential of a neuron?

I. Passive transport
II. Active transport
III. Membrane selective permeability

- A. II only [10%]
- B. III only [3%]
- C. I and III only [25%]
- ✔ D. I, II, and III [60%]

Correct
60% answered correctly

Explanation:



The unequal concentration of charged ions between the extracellular and intracellular fluid of all living cells results in an **electrochemical gradient** across the membrane that determines the membrane potential (voltage difference). The resting membrane potential of neurons is due primarily to the high concentration of potassium ions (K^+) and the low concentration of sodium ions (Na^+) inside the cell compared to the outside. The following mechanisms contribute to maintenance of the **resting membrane potential**:

- Protein channels in the cell membrane enable certain ions to move down their concentration gradient across the membrane. For example, in resting neurons, potassium (K^+) leak channels help maintain the membrane potential by enabling the passive transport (*without* using energy) of K^+ *out of* the cell. The membrane is *more* permeable to K^+ than to Na^+ (ie, **selective permeability**) due to the presence of a *greater* number of K^+ leak channels. Therefore, the resting membrane potential of neurons is approximately −70 mV, which is close to the negative equilibrium potential of K^+ **(Numbers I and III)**.

- Active transport pumps embedded in the outer membrane of neurons hydrolyze **adenosine triphosphate** (ATP) to provide energy to transport molecules against their concentration gradient. For example, sodium-potassium pumps (Na^+,K^+-ATPase) transport 2 K^+ into the cell for every 3 Na^+ moved out. This is important for maintaining the *unequal* concentration of ions across the membrane; without active transport pumps, leakage of ions through the cell membrane would eventually result in equilibration and a membrane potential of 0 mV **(Number II)**.

Educational objective:
Protein channels in the cell membrane allow passive transport of certain ions down their electrochemical gradient. This selective

membrane permeability is responsible for generating the resting membrane potential in nerve and muscle cells. Active transport pumps help maintain the concentration gradient and are critical for maintaining the resting membrane potential.

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